

Addis Ababa Science and Technology University

College of Engineering

IETP 4115- Integrated Engineering Team Project

IETP Group 3

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INTEGRATED ENGINEERING TEAM PROJECT (IETP4115)
SEMESTER 1, ACADEMIC YEAR 2026/27

AQUAPAVE

Permeable Wash-Pad Pavement for Treating and Reusing Garage Wastewater

Consolidated Project Documentation: Section 1 (Introduction), Section 2 (Multidisciplinary Project Planning), Section 3 (Data Gathering & Technical Analysis), and Section 4 (Compliance to Sustainable Development Goals)

Project Team	Group 3 — Multidisciplinary Engineering Team (7 Engineering Disciplines)
Project Advisor	Belay Gemeda
Course & Term	IETP4115 — Semester 1, 2026/27
Target Context	Commercial Automotive Garages & Car-Wash Stations, Addis Ababa, Ethiopia

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SECTION 01

INTRODUCTION

Project Background, Problem Formulation, Research Objectives, and Engineering Methodology

1.1 Background of the Project

Addis Ababa is growing rapidly: built-up urban land expanded from 42.5% to 62.6% between 2005 and 2025 [1], while the municipal water utility meets only about 40% of the city's total demand, with continuous water rationing in place since 2019 [2]. Commercial garages, automotive maintenance workshops, and car-wash stations place substantial pressure on this limited supply, since washing a single vehicle consumes between 100 and 200 liters of clean water [3].

Their wash water carries motor oil and grease, commercial soap, road dirt, and toxic heavy metals such as lead and zinc [3, 4]. In Addis Ketema Sub-City, 100% of examined garages let untreated effluent flow directly into open roadside storm drains [5], and in Jimma, discharge from car-wash stations recorded a mean turbidity of about 483 NTU and total suspended solids (TSS) of about 561 mg/L [6]. Ordinary gravity skimming tanks and simple oil-grease traps fail to clean this water adequately because detergents emulsify oil droplets [7]. Dumped garage waste also clogs drains and lowers their hydraulic capacity [8], while sealed urban surfaces have nearly doubled estimated surface runoff from 0.54 million m^3/day to 1.03 million m^3/day [1].

Permeable pavement allows water to soak through open surface voids into engineered sub-base layers that filter contaminants and retain oil: 98.7% of applied mineral oil was retained over five years in UK experimental tests [9], and a permeable pavement integrated with a built-in gravity separator successfully handled large hydrocarbon shock loads [10]. A drainage study of Addis Ketema explicitly recommended permeable pavements among its priority flood-control measures [11]. However, because standalone porous asphalt pavement did not significantly remove dissolved metals and polycyclic aromatic hydrocarbons (PAHs) [12], the concept must be tested with vehicle wash water and combined with dedicated oil separation and adsorbent filter layers. Treated water can then be stored for non-potable reuse, aligning with the 'fit for purpose' approach to urban water management [2].

> **Core Engineering Concept — AquaPave Wash-Pad Treatment Train**
>
> AquaPave replaces conventional impermeable concrete wash aprons with an active, gravity-driven treatment unit: Raw Wash Water → Pervious Concrete Slab (15–35% voids) → Choker & Non-Woven Geotextile → Adsorbent Filter Beds (Sand, Charcoal, Pumice) → Perforated Underdrain → Baffled Gravity Oil Separation Chamber → Non-Potable Reuse Tank (with ESP32 Sensor Monitoring).

1.2 Problem Statement, SDG Alignment & Research Question

1.2.1 Statement of the Problem

Garages and car-wash areas in Ethiopian cities release oily, soapy, and sediment-heavy wastewater into open drains with little or no treatment [5, 7]. This practice pollutes urban waterways and rivers, squanders scarce potable water in a city that already meets only part of its demand, and contributes directly to blocked storm drains and localized pluvial flooding [8]. Currently, there is no low-cost, locally buildable engineering system that combines a permeable wash pad, gravity oil separation, multi-media filtration, and non-potable reuse storage with real-time performance monitoring.

1.2.2 Impact on the Sustainable Development Goals (SDGs)

Untreated garage wastewater directly impedes progress toward the United Nations 2030 Agenda for Sustainable Development [13]. Implementing an on-site permeable treatment and recycling wash pad directly advances three core infrastructure and environmental goals:

- **SDG 6 — Clean Water and Sanitation (Targets 6.3 & 6.4):** Target 6.3 calls for improving water quality by reducing pollution, halving untreated wastewater, and increasing safe water recycling and reuse. Untreated garage effluent is a major urban pollution source [5, 6], and AquaPave demonstrates decentralized on-site treatment and recovery. Target 6.4 calls for substantially increasing water-use efficiency. Reusing treated wash water directly lowers municipal tap water demand: a small garage washing 20 vehicles per day at 100–200 L each consumes 2,000 to 4,000 L/day, and recycling half of that volume saves 1,000 to 2,000 L of potable water daily.
- **SDG 11 — Sustainable Cities and Communities (Target 11.5):** Target 11.5 aims to reduce the adverse impacts of water-related disasters such as urban floods. Surface runoff in Addis Ababa has nearly doubled over twenty years [1], and dumped oily sludge blocks storm drains [8]. A permeable pavement that allows wash water and rainfall to infiltrate while enabling simple surface maintenance substantially reduces hydraulic loads on municipal drains.
- **SDG 9 — Industry, Innovation and Infrastructure (Target 9.1):** Target 9.1 focuses on developing resilient, reliable, and sustainable infrastructure. AquaPave delivers a low-cost, modular pavement-based unit that small garages can construct from locally available materials and maintain without specialized machinery.

1.2.3 Central Research Question

> **Research Question**
>
> How can a low-cost permeable wash-pad pavement treat garage wastewater in Addis Ababa so that it can be reused for non-potable purposes, and how can its hydraulic and water-quality performance be reliably monitored?

System Architecture Overview: Wash water discharged onto the sloped apron infiltrates through the porous pervious concrete zone, passes downward through stratified filter layers, enters a perforated underdrain, flows slowly through a baffled oil separation chamber, and collects in a non-potable reuse container, while an instrumentation node tracks drain time, water level, and effluent clarity.

1.3 Objectives, Semester Time Frame, and Study Scope

1.3.1 General Objective

To design, construct, and experimentally evaluate a permeable wash-pad pavement system capable of treating garage wastewater generated in Addis Ababa, thereby mitigating urban environmental pollution and enabling the on-site reuse of recovered water for non-potable purposes (aligned with SDG 6, Target 6.3).

1.3.2 Specific Objectives, Timeline, and Success Indicators

Each specific objective is mapped to corresponding SDG targets, assigned a semester schedule window, and paired with quantitative success indicators (proposed indicators will be confirmed with the advisor following initial laboratory trials):

Obj.	Specific Objective Description	SDG Target	Planned Time	Success Indicator (Proposed)
1	Characterize the wastewater: Analyze baseline garage wash wastewater in a selected Addis Ababa study area (pH, turbidity, TSS, O&G, COD) within lab limits and prepare representative synthetic wash water.	SDG 6.3	November 2026	Synthetic wash water close to typical published values (~190 mg/L TSS, ~190 NTU turbidity, ~30 mg/L O&G [3]).
2	Design and build the treatment system: Design and construct a scaled prototype of a permeable concrete wash pad with multi-layered sub-base filter media, underdrain, oil separator, and reuse container.	SDG 6.3, 9.1, 11.5	Nov – Early Dec 2026	Prototype that passes a 1-hour leak test and routes all applied water through the slab without edge bypass.
3	Evaluate treatment efficiency: Measure pollutant removal efficiency comparing influent vs. underdrain/separator effluent and benchmarking runoff/quality against a conventional dense concrete control slab.	SDG 6.3	December 2026	At least 80% reduction in turbidity (or clear visual improvement), no visible effluent oil film, and much less runoff than control.

Obj.	Specific Objective Description	SDG Target	Planned Time	Success Indicator (Proposed)
4	Assess reuse potential: Determine whether treated effluent meets non-potable quality limits for on-site recycling (vehicle washing or dust control) and quantify daily municipal water savings.	SDG 6.3, 6.4	December 2026	Statement of suitable non-potable reuses compared with published limits plus estimated daily water saved per garage (L/day).
5	Assess feasibility: Evaluate structural and operational feasibility within Addis Ababa's construction material supply, local regulations, and maintenance realities (clogging and cleaning recovery).	SDG 9.1, 11.5	December 2026	Written assessment of cost, materials, construction steps, clogging progression, and post-cleaning infiltration recovery %.

1.3.3 Semester Time Frame and Achievability

All five specific objectives are structured to fit within the Semester 1 (2026/27) academic calendar. Literature synthesis, design sizing, and material acquisition take place in November; prototype fabrication, curing, and experimental testing occur across late November and December; and the live prototype demonstration is scheduled for 30 December 2026. Execution within this timeframe is achievable for four key reasons:

- **Compact Bench-Scale Prototype:** Keeping the experimental unit at a manageable bench scale enables rapid assembly, controlled hydraulic dosing, and triplicate testing within a few weeks.
- **University Laboratory Resource Availability:** Core aggregates, washed sand, charcoal, pumice, cement, glassware, and basic tools are readily available in university laboratories and local markets, minimizing procurement delays and cost.
- **Front-Loaded Critical-Path Tasks:** Time-sensitive steps—specifically trial pervious concrete mixing, casting, and 7-day wet curing—are initiated first in Week 5, with modular permeable pavers designated as an immediate backup if the cast slab underperforms.
- **Sensor & Instrument Redundancy:** Experimental validation does not rely on any single electronic sensor or analytical meter; calibrated measuring jugs, stopwatches, and photographic turbidity comparisons can replace automated instruments if needed.

1.3.4 Scope of the Study (In-Scope vs. Out-of-Scope)

In-Scope Activities & Deliverables	Out-of-Scope Boundaries & Exclusions
• Prototype: A scaled bench-top unit comprising a permeable concrete wash zone, multi-layer sub-base filter bed, perforated underdrain, baffled oil separation chamber, and reuse storage tank.	• Full-scale commercial installation: Only one bench-scale prototype will be built; a full-scale outdoor garage wash pad would also require roofing or a stormwater diverter.
• Experimental Testing: Controlled runs using synthetic wash water (and real garage grab samples where feasible), compared against a dense concrete control slab, including soil clogging and surface cleaning tests.	• Potable water treatment: Treated water is strictly designated for non-potable reuse applications only, and laboratory test results do not constitute formal regulatory certification.
• Performance Monitoring: Manual volumetric and drain-time measurements supplemented by an ESP32 microcontroller with water-level and turbidity sensors and maintenance alerts.	• Guaranteed removal rates: Numerical removal targets are experimental goals to be tested; emulsified oils, surfactants, and dissolved pollutants may only be partially removed.
• Technical Documentation: Complete engineering report, architectural/system drawings, bill of materials, and empirical test datasets.	• Long-term field durability: Only short-term accelerated clogging is evaluated within the single-semester academic window.

1.4 Design Approach, Engineering Methodology, Equations & Risk Matrix

1.4.1 Overall Five-Step Methodology

The project follows a structured Design–Build–Test–Refine engineering workflow in five steps aligned with Objectives 1 through 5:

- **Step 1 — Plan & Characterize (Objective 1):** Review technical literature, visit a local Addis Ababa garage if accessible, and define baseline wastewater characteristics and design requirements.
- **Step 2 — Design & Select Subsystems (Objective 2):** Select tray dimensions, pavement mix proportions, filter media layers, pipe diameters, oil separator baffles, and sensors matching laboratory availability.
- **Step 3 — Build Prototype (Objective 2):** Fabricate the frame, cast and cure the pervious and control slabs, pack the filter media, plumb the underdrain/separator, and wire the ESP32 node.
- **Step 4 — Test & Compare (Objectives 3 & 4):** Dose synthetic wash water onto both the AquaPave prototype and dense control slab, collect stage-wise samples, and measure hydraulic and water-quality metrics.
- **Step 5 — Improve & Report Feasibility (Objective 5):** Resolve operational issues, evaluate clogging/cleaning behavior, calculate water savings and payback, and compile the final technical report.

Integrating physical straining, multi-media adsorption, and gravity oil separation follows established car-wash treatment literature [3]. In comparable laboratory studies, combining sand-and-gravel filtration with activated carbon removed approximately 90% of suspended solids and 86% of oil and grease [4].

1.4.2 How the Integrated System Works

- **1. Wash Surface & Permeable Slab:** Wash water flows across a gently sloped apron and infiltrates through the open interconnected pores of the pervious concrete slab, which traps coarse road grit and debris.
- **2. Multi-Layer Sub-Base Filtration:** Geotextile fabric, washed sand, wood charcoal/activated carbon, and volcanic pumice filter fine suspended solids and adsorb emulsified oils, surfactants, and heavy metals.
- **3. Underdrain & Oil Separation Chamber:** A perforated PVC pipe collects percolated water and routes it slowly through a baffled gravity chamber where free oil droplets float to the surface and are skimmed off.
- **4. Non-Potable Reuse Container:** Clarified effluent overflows into a collection tank for non-potable recycling (e.g., vehicle pre-rinsing or floor washing).
- **5. Monitoring & Clogging Alert:** Water level, drain time, and clarity are recorded manually and logged via an ESP32 microcontroller that triggers a maintenance alert when drainage slows or the tank fills.

1.4.3 Core Engineering Design Considerations

- **Hydraulics & Anti-Bypass Sealing:** The slab infiltration rate must exceed the applied dosing rate by at least 5x to prevent surface overflow. The approach apron slopes at 2% to 5% toward the permeable zone, and all edges are sealed with a waterproof liner so water cannot bypass the filter media.
- **Pervious Concrete Mix (ACI 522R-10):** Eliminates fine sand, uses single-size coarse aggregate, maintains a water-to-cement ratio of 0.27–0.34, and targets 15%–35% void content [14]. Verified by the hand-squeeze test (cohesive glossy ball without dripping paste), placed without over-compaction, and cured under plastic for 7 days. Permeable pavers with gravel joints serve as the backup.
- **Gravity Oil Separation (Stokes' Law):** Droplet rise velocity follows $v = [g \cdot d^2 \cdot (\rho_w - \rho_o)] / (18 \cdot \mu)$, with required surface area $A = Q / v$ [15]. For a 100 µm oil droplet rising at ~0.8 mm/s at Q = 2 L/min, minimum surface area is ~0.04 m^2 (idealized, as detergents reduce coalescence).

1.4.4 Materials, Test Procedure, and Synthetic Wash Water Recipe

Required Materials & Equipment: Portland cement, single-size gravel, washed sand, crushed charcoal, volcanic pumice, non-woven geotextile, plastic liner sheeting, wooden/metal molds, PVC pipes and end-caps, containment tray, baffled oil separator box, reuse tank, clean silty soil, motor oil (or vegetable oil proxy), commercial vehicle detergent, measuring jugs, stopwatch, scale, sample bottles, turbidity meter/sensor, ESP32 board, and PPE (gloves, goggles, dust mask).

Seven-Step Experimental Test Procedure: (1) Run a 1-hour clean-water leak and bypass check; (2) Measure baseline clean-water drain time; (3) Prepare synthetic wash water (per 10 L water: ~1.9 g clean soil for ~190 mg/L TSS, ~0.33 mL motor oil for ~30 mg/L O&G, and ~0.65 g detergent for ~13 mg/L active surfactant, adjusted to 150–250 NTU; plus a heavy stress-test batch at 10 g soil, 3 g oil, and 3 g detergent per 10 L); (4) Pour onto the pad at a steady rate and sample influent, underdrain effluent, post-oil chamber water, and reuse tank water; (5) Repeat identical dosing on the dense concrete control slab; (6) Execute at least 3 replicates per test; and (7) Conduct an accelerated clogging test by adding fine soil incrementally, recording drain-time rise, then brushing/rinsing the slab to measure permeability recovery.

1.4.5 Quantitative Data Analysis Equations

Parameter / Index	Mathematical Formulation	Guiding Standard
Infiltration Rate (I)	$I = V / (A_{slab} \times t)$	ASTM C1701 / C1701M [16]
Runoff Coefficient (C_r)	$C_r = V_{runoff} / V_{applied}$	Surface Water Balance
Void Content (φ)	$\varphi\ (\%) = [1 - ((M_{dry} - M_{sub}) / (\rho_w \times V_{bulk}))] \times 100$	ASTM C1754 / C1754M [17]
Removal Efficiency (η)	$\eta\ (\%) = [(C_{in} - C_{out}) / C_{in}] \times 100$	Treatment Performance
Clogging Ratio (R_c)	$R_c = t_{drain(current)} / t_{drain(baseline)}$	Maintenance Assessment

1.4.6 Project Risk Matrix and Safety Protocols

Risk Event	Severity	Mitigation & Contingency Action
Cast slab cracks or drains too slowly	High	Cast early trial mixes in Week 5; switch to modular permeable pavers with gravel joints as backup.
Water bypasses around slab edges	High	Seal perimeter edges with flexible waterproof liner and verify via 1-hour leak test before every run.
Lab meters or ESP32 sensors unavailable	Medium	Use calibrated measuring jugs, stopwatches, and standardized visual/photographic clarity comparisons.
Detergent emulsifies oil droplets	Medium	Document quantitatively as an engineering finding and add an activated charcoal polishing layer if time permits.
Schedule compression in December	Medium	Keep prototype compact (bench-scale), start 7-day concrete curing early, and parallelize subsystem builds.

Laboratory Safety & Environmental Disposal: Gloves, goggles, and dust masks are mandatory during cement mixing and media sieving. No effluent from the prototype is ever consumed, and all oily wastewater and spent filter media are collected in sealed containers for proper lab disposal.

SECTION 02

MULTIDISCIPLINARY PROJECT PLANNING

Team Organization Across Seven Engineering Disciplines, Subsystem Interfaces & 12-Week Work Plan

2.1 Multidisciplinary Team Structure and Task Allocation

The AquaPave system combines structural pavement design, chemical filtration media, environmental wastewater evaluation, architectural spatial modeling, electromechanical fabrication, sensor electronics, and embedded software engineering. To establish clear ownership across all ten team members representing seven engineering departments, technical responsibilities are assigned as follows:

No.	Engineering Department	Assigned Responsibilities & Technical Deliverables
1	Civil Engineering	Design the pervious concrete pavement mix, determine pavement layer thicknesses, prepare the concrete mixture, cast and cure the pervious and control slabs, and conduct ASTM infiltration tests.
2	Chemical Engineering	Select and evaluate filter materials such as washed sand, wood charcoal, volcanic pumice, and activated carbon; assess their pollutant adsorption and removal capacity.
3	Environmental Engineering	Investigate Addis Ababa garage wastewater characteristics, support wastewater sampling, formulate synthetic wash water, evaluate environmental impacts, and assess non-potable water reuse potential.
4	Architectural Engineering	Prepare the overall wash-pad prototype layout, develop structural and visual 2D/3D drawings, optimize slope geometry, and build the architectural scale model.
5	Electromechanical Engineering	Design and fabricate the structural prototype frame, watertight washing tray, slotted drainage pipes, baffled oil separator, reuse collection tank, and mechanical connections.
6	Electrical & Computer Eng. (1)	Select suitable water-level and turbidity sensors, design sensor signal-conditioning circuits, and develop the electrical/solar power supply arrangement.
7	Electrical & Computer Eng. (2)	Interface the ESP32 microcontroller with the water-level, turbidity, and flow sensors, and execute hardware wiring and waterproofing integration.
8	Electrical & Computer Eng. (3)	Test and calibrate the sensor system against reference standards, troubleshoot electrical connections, and verify accurate sensor readings.
9	Software Engineering (1)	Develop the embedded ESP32 firmware for continuous sensor data acquisition, signal filtering, processing, and timestamped data logging.

No.	Engineering Department	Assigned Responsibilities & Technical Deliverables
10	Software Engineering (2)	Develop the hydraulic clogging detection algorithm, automated maintenance alert system, and user monitoring interface, supporting end-to-end software integration.

2.2 Subsystem Integration & Cross-Disciplinary Interfaces

Because AquaPave integrates civil, chemical, environmental, mechanical, electrical, and software components into a single bench-scale prototype, disciplined coordination across four primary engineering interfaces is essential to prevent dimensional mismatches or hydraulic bottlenecks:

- **Interface 1 — Civil, Chemical & Architectural Engineering (Pavement-to-Filter Coupling):** Architectural Engineering defines the wash-pad surface footprint and 2%–5% approach slope; Civil Engineering ensures the cast pervious concrete slab meets void content (15%–35%) and strength requirements without raveling; Chemical Engineering specifies the non-woven geotextile and stratified sand, charcoal, and pumice bed depths positioned directly beneath the slab.
- **Interface 2 — Chemical & Electromechanical Engineering (Filter-to-Hydraulics Coupling):** Electromechanical Engineering fabricates the watertight containment tray, slotted PVC underdrain pipe, baffled gravity oil separation chamber, and reuse collection tank to match the exact hydraulic head-loss and laminar retention time specifications required by Chemical Engineering.
- **Interface 3 — Environmental & Civil/Chemical Engineering (Water Quality & Benchmark Protocol):** Environmental Engineering establishes the synthetic wash-water formulation, stage-wise sampling schedule (influent, underdrain, oil chamber, reuse tank), and non-potable reuse limits used by Civil and Chemical engineers to evaluate treatment efficiency and clogging recovery.
- **Interface 4 — Electromechanical, Electrical & Software Engineering (Smart Monitoring Integration):** Electromechanical Engineering integrates waterproof sensor mounting ports into the separator and collection tank; Electrical & Computer Engineering wires and calibrates the ESP32 water-level and turbidity sensors; Software Engineering implements the clogging detection algorithm and maintenance alert interface.

> **Work Plan Governance Note**
>
> The following 12-week schedule represents the proposed Semester 1 (2026/27) operational roadmap. Each week pairs a primary milestone objective with concrete disciplinary tasks and remains adaptable to official IETP4115 review dates issued by Advisor Belay Gemeda.

2.3 Detailed 12-Week Implementation Schedule (Weeks 1–6)

Week 1: Team Meeting and Communication

Objective: *To meet team members and establish a clear understanding of the project.*

- Gather at the advisor's office and introduce all ten team members.
- Discuss the overarching project objectives, academic deliverables, and advisor expectations.
- Clarify the specific responsibilities of each of the seven engineering disciplines.
- Establish communication methods (messaging channels, shared files) and regular weekly meeting schedules.

Week 2: Brainstorming and Project Selection

Objective: *To generate ideas and select the most suitable multidisciplinary engineering project.*

- Individually identify possible engineering project ideas addressing local urban needs.
- Share and discuss all proposed ideas across the entire team.
- Evaluate candidate ideas based on technical feasibility, cost, available lab materials, and expected SDG impact.
- Select the AquaPave permeable wash-pad project idea in consultation with the advisor.
- Decide on the final project title and define the primary and specific project objectives.

Week 3: Proposal Preparation and Submission

Objective: *To prepare and submit the formal IETP4115 project proposal.*

- Conduct preliminary literature research on permeable pavements and garage wastewater treatment.
- Define the project problem statement, SDG alignment, general/specific objectives, and proposed solution.
- Prepare the project introduction, experimental methodology, and risk mitigation plan.
- Develop the preliminary project work plan, including task allocation across disciplines and Gantt schedule.
- Review, edit, and submit the complete proposal to the advisor.

Week 4: Intensive Planning, Design, and Material Selection

Objective: *To finalize the preliminary technical design and prepare for prototype development.*

- System Design: Develop the overall AquaPave layout; design pavement layers, filtration section depths, and wastewater collection/separator geometry; prepare preliminary block diagrams and technical drawings.
- Material Selection: Identify suitable materials for pervious concrete, sand, gravel, charcoal, and pumice; select PVC pipes, collection tanks, and frame materials; estimate the project budget and identify local suppliers.
- Electronic & Software Planning: Select water-level and turbidity sensors; plan the ESP32-based monitoring circuit; define required firmware and alert software architecture.

Week 5: Material Procurement and Pavement Preparation

Objective: *To acquire required materials and begin construction of pavement and filtration components.*

- Purchase or requisition the selected construction aggregates, cement, filter media, and electronic components.
- Prepare wooden/metal pavement molds and proportion the pervious concrete mixture (verifying via hand-squeeze test).
- Cast the pervious concrete slab and dense control slab samples and initiate the 7-day wet curing process.
- Wash, dry, and sieve the sand, gravel, charcoal, and pumice filter media.
- Conduct initial bench testing of the sensors and ESP32 electronic components.

Week 6: Filtration and Mechanical Subsystem Development

Objective: *To develop the filtration bed and mechanical containment structure of the prototype.*

- Arrange the graded filtration layers (pumice, charcoal, sand, non-woven geotextile) inside the lined containment box.
- Design and assemble the slotted PVC underdrain and wastewater collection pipes.
- Construct the structural prototype frame and sloped washing-pad tray.
- Prepare the baffled oil separation chamber and reuse collection tank and connect the drainage plumbing.
- Inspect the curing pavement samples and monitor structural hardening progress.

2.3 Detailed 12-Week Implementation Schedule (Weeks 7–12)

Week 7: Electronic and Software Subsystem Development

Objective: *To develop the sensor-based monitoring and clogging alert system.*

- Connect the ESP32 microcontroller with the selected water-level and turbidity sensors.
- Develop the regulated electrical connections and low-power supply arrangement.
- Write the initial embedded program for collecting and filtering sensor readings.
- Develop the software modules for continuous monitoring of water level and effluent turbidity.
- Begin developing the hydraulic clogging detection algorithm and maintenance alert functions.

Week 8: Prototype Assembly and System Integration

Objective: *To combine pavement, filtration, mechanical, electronic, and software subsystems into one working prototype.*

- Assemble the cured pervious concrete slab over the prepared filtration layers and seal perimeter edges.
- Install and seal the underdrain pipes, baffled oil separation chamber, and wastewater reuse collection tank.
- Integrate the waterproofed sensor probes and ESP32 enclosure with the prototype.
- Upload and test the developed data-logging and alert software on the integrated hardware.
- Check all mechanical, hydraulic, and electrical connections across subsystems and run a 1-hour leak test.

Week 9: Initial Testing and Performance Evaluation

Objective: *To evaluate the initial hydraulic and treatment performance of the AquaPave prototype.*

- Conduct baseline clean-water infiltration tests and initial synthetic wastewater flow trials.
- Observe the drainage performance and runoff reduction of the pervious slab compared to the control slab.
- Measure water turbidity, TSS, and visual oil removal before and after filtration and oil separation.
- Check the accuracy and stability of the water-level and turbidity monitoring system.
- Identify any edge leakage, media blockage, emulsified oil carryover, or sensor integration issues.

Week 10: System Improvement and Optimization

Objective: *To improve prototype performance and reliability based on initial Week 9 test results.*

- Identify and correct any hydraulic leaks, drainage bottlenecks, or structural issues observed during testing.
- Optimize the arrangement or thickness of the charcoal/pumice filtration layers if necessary.
- Adjust the sensor calibration curves and software clogging alert thresholds.
- Improve the wastewater drainage baffles and reuse collection piping.
- Repeat selected experimental tests to verify performance improvements.

Week 11: Final Testing, Data Analysis, and Documentation

Objective: *To verify overall system performance in triplicate and prepare final project documentation.*

- Conduct final triplicate infiltration, filtration, and accelerated soil-clogging/cleaning performance tests.
- Evaluate the wastewater treatment efficiency and non-potable reuse collection results.
- Analyze and interpret all collected experimental data (means, spreads, removal efficiencies, water savings).
- Prepare clean tables, performance graphs, prototype photographs, and technical results.
- Write, edit, and review the comprehensive final project report.

Week 12: Final Presentation and Project Submission

Objective: *To present the completed AquaPave project and submit all required final documentation.*

- Complete the final project report and presentation slides.
- Prepare the physical AquaPave prototype and synthetic wash-water batch for live demonstration.
- Rehearse the project presentation with all ten team members.
- Present the project results and live prototype demonstration to the advisor and evaluation panel (30 Dec 2026).
- Submit the final report and required project materials (1 Jan 2027).

2.4 Project Milestone Summary & Deliverable Gates

To track schedule adherence across all seven engineering departments, six formal milestone gates govern the 12-week execution cycle:

Milestone Gate	Schedule Window	Lead Disciplines	Required Verification Output
Gate 1: Proposal Approval	Week 3 (Nov 2026)	All 7 Disciplines	Submitted and advisor-approved IETP4115 Project Proposal & 12-Week Plan.
Gate 2: Concrete Casting & Curing	Week 5 (Nov 2026)	Civil & Architectural Eng.	Cast pervious concrete slab and dense control slab entering 7-day wet cure.
Gate 3: Progress Report	Week 6 (4 Dec 2026)	Environmental & Chemical Eng.	Documented wastewater characterization and verified synthetic wash-water recipe.
Gate 4: Integrated Prototype	Week 8 (Mid Dec 2026)	Electromech., ECE & Software	Fully assembled, leak-tested prototype with active ESP32 sensor logging.
Gate 5: Live Demonstration	Week 12 (30 Dec 2026)	All 7 Disciplines	Live treatment demonstration achieving ≥80% turbidity reduction and zero oil sheen.
Gate 6: Final Submission	Week 12 (1 Jan 2027)	All 7 Disciplines	Complete Final Engineering Report, feasibility/payback analysis, and SDG compliance audit.

> **Critical-Path Buffer Management**
>
> Because Portland cement hydration requires a minimum 7-day wet cure (Week 5 to Week 6) before structural loading or infiltration testing, Civil Engineering initiates trial batching in early Week 5 while Electromechanical and Electrical teams fabricate the frame, separator box, and sensor harness concurrently. If any cast slab exhibits inadequate permeability during Week 6 inspection, pre-sourced modular permeable concrete pavers are immediately substituted so Week 8 integration stays on schedule.

SECTION 03

DATA GATHERING AND TECHNICAL ANALYSIS

Urban Hydrological Stress, Garage Wastewater Characterization, Permeable Pavement Mechanics & Design Synthesis

3.1 Preliminary Information: Behaviour Trends & Wastewater Characterization

3.1.1 Urban Context and Hydrological Stress in Addis Ababa

Addis Ababa is currently undergoing unprecedented, rapid urbanization. Spatial analysis indicates that the city's built-up land area expanded from 42.5% in 2005 to 62.6% in 2025 [1]. This intensive sealing of natural soil surfaces with conventional impermeable asphalt and concrete has severely altered the regional water balance. Natural rainfall infiltration has been drastically diminished, while surface runoff volumes rose from an estimated 0.54 million m^3/day in 2005 to 1.03 million m^3/day in 2025 [1, 12].

Simultaneously, the city faces a severe water supply crisis. Estimates from water resource specialists indicate that the Addis Ababa Water and Sewerage Authority (AAWSA) is capable of meeting only about 40% of the city's total potable water demand, which stands at approximately 1.2 million m^3/day [2]. Consequently, municipal water rationing has been continuously in effect since 2019, and overreliance on groundwater extraction has depleted local aquifers to alarming depths [2]. The resulting deficit forces both residential populations and commercial enterprises to operate under intermittent water availability.

3.1.2 Growth of Automotive Washing Operations and Water Consumption

Parallel to rapid urban growth, commercial vehicle maintenance workshops, auto-repair garages, and informal car-wash stations have proliferated across all sub-cities of Addis Ababa [4, 8]. Vehicle washing is an inherently water-intensive operation: standard industry estimates show that washing a single light vehicle consumes between 100 and 200 liters of water [3]. When these commercial entities utilize high-grade municipal tap water for vehicle cleaning, they directly deplete public drinking reserves and exacerbate local municipal water stress [2, 3].

3.1.3 Baseline Characterization of Garage and Car-Wash Wastewater

Field investigations and water quality monitoring across Ethiopian cities (such as Addis Ketema Sub-City in Addis Ababa and Jimma) demonstrate that automotive service stations represent major stormwater hot-spots [4, 5, 15]. Wash-pad runoff carries heavy physical, chemical, and biological pollutant loads generated from vehicle exterior cleaning, engine degreasing, and mechanical repairs [3, 4, 6]:

- **Total Suspended Solids (TSS):** Mean TSS concentrations range from 186 mg/L up to 561 mg/L (with median literature values around 186–187 mg/L) [3, 5]. This load consists of fine road dust, engine grit, soil particles, and worn brake lining residue [3, 6].
- **Turbidity:** Discharged wash water exhibits extreme cloudiness, with turbidity values ranging between 187 NTU and 483 NTU [3, 5].
- **5-Day Biochemical Oxygen Demand (BOD5) & COD:** Effluents show high organic enrichment, with mean BOD5 reaching 334 mg/L and elevated COD values, driven by organic grease, food waste, dirt, and biodegradable surfactants [4, 5, 8].
- **pH Levels:** Raw wash water typically exhibits a pH range of 6.5 to 8.5, fluctuating depending on the specific soap formulations and cleaning solvents applied [3, 6].
- **Hydrocarbons, Detergents, and Heavy Metals:** Oil and grease concentrations span from 12 mg/L to 325 mg/L, with standard median values reported at 28 mg/L, including used motor oils, diesel fuel, gasoline, hydraulic fluids, and lubricating greases [3, 15, 16]. Commercial detergents introduce high levels of anionic surfactants (median 13 mg/L) and elevated orthophosphates [3, 5], while vehicle wear releases Lead (Pb = 0.53 mg/L), Zinc (Zn = 1.98 mg/L), Copper (Cu), and Cadmium (Cd) [6, 15].

Water Quality Parameter	Reported Range / Mean Value	Primary Source
Total Suspended Solids (TSS)	186.0 mg/L – 561.0 mg/L [3, 5]	Road grit / soil
Turbidity	187.0 NTU – 483.0 NTU [3, 5]	Suspended fines
Biochemical Oxygen Demand (BOD5)	334.0 mg/L [4, 5]	Organics / soaps
Oil and Grease (O&G)	12.0 mg/L – 325.0 mg/L [3]	Lubricants / fuel
Anionic Surfactants	13.0 mg/L (Median) [3, 5]	Detergents
Lead (Pb)	0.53 mg/L [6]	Exhaust / batteries
Zinc (Zn)	1.98 mg/L [6]	Tire wear / galvanized parts

3.1.4 Environmental and Public Health Impacts of Current Discharge Practices

In Addis Ababa and secondary Ethiopian cities, 100% of surveyed small garages dump their liquid waste directly into unlined roadside ditches and storm channels [4, 11]. This practice produces four cascading environmental and public health hazards:

- **1. Hydraulic Drain Blockage and Urban Flooding:** Waste oil mixed with fine sediment forms a viscous sludge that settles in urban drains, drastically reducing their hydraulic capacity [11]. In Addis Ketema, blocked inlets and undersized pipes loaded with garage sediment directly cause localized urban pluvial flooding [1, 12].
- **2. Ecological Damage to Surface Water:** 1 liter of waste oil can contaminate up to 1,000,000 liters of clean water [16]. When wash water reaches receiving rivers like the Little Akaki or Shankela, oil forms a micro-thin surface film that blocks sunlight and prevents atmospheric oxygen dissolution, suffocating fish, aquatic plants, and benthic microbes [4, 16]. Elevated orthophosphates trigger severe eutrophication and algal blooms [5].
- **3. Toxicological and Human Health Hazards:** Waste oil contains hazardous Polycyclic Aromatic Hydrocarbons (PAHs), benzene, toluene, and heavy metals [16]. Exposure causes contact dermatitis, oil acne, lipid pneumonia, and long-term carcinogenic risks (squamous cell carcinoma, bladder, and liver cancers) [16].
- **4. Persistence in Aquatic Sediments:** Environmental records (such as the 2010 Gulf of Mexico Deepwater Horizon spill assessment) demonstrate that hydrocarbon sediments in aquatic systems remain 10 times higher than baseline even 8 years post-contamination, permanently disrupting benthic feeding and breeding cycles [16].

3.1.5 Limitations of Conventional Primary Treatment Systems

Small commercial garages often rely on basic gravity skimming tanks or simple oil-grease interceptors [8]. However, field evaluations show these units are inadequate for vehicle wash pads: gravity oil-water separators (such as API separators) rely on density differences to float free oil droplets [8, 15], but when detergents and surfactants are present, they emulsify oil droplets into microscopic dispersions, reducing gravity skimming efficiency to under 30%–40% [3, 8]. Moreover, standard skimming tanks fail to filter fine TSS and cannot remove dissolved heavy metals or ionic surfactants [3, 8].

3.1.6 Formulation of Synthetic Wash Water for Bench-Scale Evaluation

To enable repeatable, controlled laboratory testing of the AquaPave prototype under standard conditions, a synthetic wash-water matrix is formulated based on mean parameter values documented across literature [3, 5, 6]:

Target Parameter	Standard Laboratory Additive	Dosed Quantity
Total Suspended Solids (TSS)	Local silty-clay soil & road grit	350 – 500 mg/L
Oil and Grease (O&G)	Used 15W-40 engine oil + diesel	100 – 150 mg/L

Target Parameter	Standard Laboratory Additive	Dosed Quantity
Surfactants	Commercial vehicle wash soap	15 – 25 mg/L
Heavy Metals (Proxy)	Zinc Nitrate / Lead Nitrate salts	2.0 / 0.5 mg/L
Base Solvents	Tap water baseline (pH approx. 7.2)	Balance

3.2 Technical Documentation: Permeable Pavements, Media & Clogging

3.2.1 Structural, Hydrologic, and Hydraulic Performance of Permeable Pavements

Permeable Pavement Systems (PPS) function as dual-purpose infrastructure: providing structural support for traffic loads while simultaneously serving as an on-site stormwater drainage, storage, and filtration media [13, 14, 18, 19].

Surface Permeability and Infiltration Mechanics: Pervious concrete (PC) and modular permeable interlocking concrete pavers (PICP) feature surface void spaces ranging from 15% to 25% [14, 18, 19]. Surface infiltration rates for newly constructed, clean permeable pavements are exceptionally high, typically spanning 120 cm/h to over 4,000 cm/h [18, 19]. These rates far exceed natural peak precipitation intensities and heavy wash-pad application flows, preventing surface runoff generation under normal operational conditions [18, 19].

Pavement System Type	Surface Infiltration Rate Range	Reference
Visibly Clean Pervious Concrete (PC)	2,000 cm/h – 4,000 cm/h	Bean et al. [19]
Visibly Clean PICP	100 cm/h – 2,000 cm/h	Pratt et al. [19]
Porous Asphalt (PA)	111 cm/h – 635 cm/h	Hou et al. [19]
Clogged Permeable Surface	0.0 cm/h – 16 cm/h	Bean et al. [19]
Conventional Dense Asphalt	0.0 cm/h (100% Runoff)	Control Baseline

Hydrologic Attenuation and Life Cycle Cost Dynamics: Comparative monitoring between permeable pavements and dense, impermeable controls highlights peak discharge reductions of 70% to 98%, complete volume attenuation for small-to-moderate inputs, and hydrograph outflow lag times ranging from 28 minutes to over 21 hours [18, 19]. Life Cycle Assessment (LCA) studies indicate that raw material acquisition and cement production account for 70% to 92% of life-cycle energy and GHG emissions, and initial construction costs can be 17.1% to 31.5% higher than conventional pavements [13]. However, these upfront costs are offset by reducing downstream stormwater piping [13, 14], eliminating separate

centralized settling basins [13, 18], mitigating Urban Heat Island (UHI) temperatures via evaporative cooling [13, 21], and harvesting non-potable water on-site [10, 13].

3.2.2 Pollutant Retention, Treatment Mechanisms, and Media Selection

Permeable Pavement Systems achieve water quality restoration through physical, chemical, and biological mechanisms operating within the surface layer, geotextile interface, and sub-base aggregate matrix [13, 14, 18, 19]:

- **Physical Filtration of Suspended Solids:** Coarse particulate matter is intercepted at the pavement surface via pore straining, while fine particulates are trapped within the top bedding aggregate and non-woven geotextile layers, achieving 80% to 96% TSS removal [14, 18, 19].
- **Microbial Hydrocarbon Degradation (In-Situ Bioreactors):** Naturally established microbial populations (fungi, bacteria, testate amoebae, and ciliates) colonize sub-base aggregates and geotextiles, retaining >98.7% of applied mineral oil and aerobically biodegrading ~45% within 6 months without artificial inoculation [14, 22].
- **Chemical Heavy Metal Immobilization & pH Buffering:** Exfiltrate passing through cementitious pervious concrete experiences alkaline leaching, elevating outflow pH to 8.0–9.5 [18, 19]. This alkaline environment precipitates dissolved Lead (Pb), Zinc (Zn), Copper (Cu), and Cadmium (Cd) as insoluble hydroxides and carbonates, achieving 74%–93% Pb removal and 55%–96% Zn removal [18, 19].
- **Sub-Base Adsorbent Enhancement:** Because surface pervious concrete alone has limited capacity for dissolved surfactants and emulsified oils [13, 18], AquaPave integrates activated charcoal/carbon (for organic/surfactant adsorption) and local volcanic pumice/sand (for cation exchange), achieving ~90% TSS and ~86% O&G removal in dual-stage filtration beds [6, 13, 14].

Pollutant Category	Concentration / Mass Reduction %	Reference
Total Suspended Solids (TSS)	80.0% – 96.0%	[14, 18, 19]
Oil and Grease / Motor Oil	98.0% – 99.0%	[14, 22]
Lead (Pb)	74.0% – 93.0%	[18, 19]
Zinc (Zn)	55.0% – 96.0%	[18, 19]
Copper (Cu)	52.0% – 86.0%	[18, 19]
Total Phosphorus (TP)	40.0% – 84.0%	[13, 18]
Total Nitrogen (TN)	15.0% – 61.8% (Variable)	[13, 18]

Filter Layer Material	Target Pollutant Class	Dominant Mechanism
Coarse Aggregates (10–20 mm)	Large debris, hydraulic storage	Void retention
Clean Fine Stone (2–5 mm)	Coarse sediment, TSS	Mechanical sieve
Non-Woven Geotextile	Fine TSS, free oil slicks	Biofilm & micro-filtration
Activated Charcoal / Carbon	Emulsified oils, surfactants	Physical-chemical adsorption

Filter Layer Material	Target Pollutant Class	Dominant Mechanism
Volcanic Pumice / Sand	Dissolved metals (Pb, Zn)	Cation exchange

3.2.3 Operational Challenges: Clogging Mechanics and Climate Resilience

Physical Clogging Dynamics: Clogging represents the single largest operational threat to long-term permeable pavement functionality [13, 18, 19]. The process follows five sequential stages: (1) Deposition of road soil, wash grit, and organic matter on the surface; (2) Migration of fine particles (sediment-to-pore diameter ratio < 0.1) into upper voids; (3) Lodging of sediments within the top 1.0 to 2.0 cm layer of the pervious matrix; (4) Binding of fine dust by detergent soap and oil residues into an impermeable crust; and (5) Surface infiltration dropping from >1,000 cm/h down to <1.5 cm/h [13, 18, 19]. Traffic loads worsen clogging by grinding surface dust into pores [18, 19].

Rejuvenation Protocols and Restoration Efficiency: To sustain long-term hydraulic capacity, periodic surface maintenance is required [13, 18, 19]:

Rejuvenation Method	Infiltration Restoration %	Operational Assessment & Reference
Dry Vacuum Sweeping	15.0% – 35.0% (Surface)	Removes loose litter; low efficiency on compacted crusts [18, 19]
High-Pressure Washing Only	50.0% – 70.0% (Risk deep)	Dislodges top 2 cm crust, but may push fines deeper [13, 18]
Combined Pressure + Vacuum	80.0% – 95.0% (Effective)	Optimal rehabilitation method restoring near-initial rates [18, 19]
Manual Top 25 mm Sand Replace	60.0% – 85.0% (PICP)	Effective manual maintenance option for block pavers [18]

Freeze-Thaw, De-Icing Salts, and Thermal Behavior: In cold regions, open-graded sub-bases prevent frost heave by draining rapidly, though winter road salts (NaCl, CaCl2) introduce chloride ions that can mobilize bound heavy metals via cation replacement [18, 19]. For tropical highland applications like Addis Ababa, road salting is completely absent, eliminating chloride mobility concerns. Furthermore, permeable pavements mitigate the Urban Heat Island (UHI) effect through daytime evaporative cooling of stored sub-base moisture [13, 21].

3.2.4 Secondary Water Quality Considerations: Microorganisms

While indigenous protozoan and bacterial populations (flagellates, ciliates, gymnamoebae) effectively biodegrade hydrocarbons, permeable pavements do not consistently eliminate bacterial pathogens (e.g., E. coli, total coliforms) introduced by organic wash debris [14, 18]. If treated effluent is intended for high-contact non-potable spray reuse, secondary disinfection (such as simple chlorination or solar UV exposure) may be incorporated [14, 18].

3.3 Synthesis: Non-Potable Reuse Targets & Integrated AquaPave Concept

3.3.1 Non-Potable Water Quality Reuse Targets

To quantify the treatment success of the AquaPave prototype, underdrain and oil-separator effluent quality is evaluated against established vehicle washing water recycling targets [3] and provisional Ethiopian industrial discharge limits for interceptor outflows [18]:

Water Quality Parameter	Non-Potable Car-Wash Reuse Target	Provisional Ethiopian Standards
Total Suspended Solids (TSS)	< 50.0 mg/L	< 50.0 mg/L
Turbidity	< 20.0 NTU	N/A
Oil and Grease (O&G)	< 5.0 mg/L	< 15.0 mg/L
pH	6.5 – 8.5	6.0 – 9.0
Lead (Pb)	< 0.1 mg/L	< 0.1 mg/L
Zinc (Zn)	< 1.0 mg/L	< 2.0 mg/L

3.3.2 Integrated Architectural and Engineering Concept for AquaPave

Synthesizing all hydrological, structural, chemical, and environmental findings gathered across Section 3, the AquaPave wash-pad system consolidates surface infiltration, multi-media adsorption, gravity oil separation, and non-potable water harvesting into a single compact unit:

> **Integrated AquaPave System Flow Architecture**
>
> Raw Washwater Inflow → Surface Layer (Pervious Concrete, ~20% Void Content) → Choker Course (5–10 mm Aggregate) → Non-Woven Geotextile Layer → Sub-Base Adsorbents (Washed Sand, Activated Charcoal, Volcanic Pumice) → Perforated PVC Underdrain → Gravity Oil Separation Baffle Chamber → Non-Potable Reuse Storage Tank.

This sequential multi-barrier configuration directly addresses the failure modes of conventional garage oil traps: coarse solids are captured at the concrete surface, fine silts are strained by the geotextile, emulsified hydrocarbons and surfactants are adsorbed onto activated charcoal, dissolved heavy metals are immobilized by alkaline pH buffering and pumice cation exchange, and residual free oil droplets are skimmed in the baffled gravity chamber prior to storage.

SECTION 04

COMPLIANCE TO REQUIREMENTS (SDGs)

Alignment with the 2026/27 IETP Sustainable Development Goals, Verification Evidence & Traceability

4.1 Purpose and Tiered Compliance Framework

This section demonstrates the compliance of the AquaPave project with the Sustainable Development Goals (SDGs) designated for the 2026/27 IETP projects: SDG 1 (No Poverty), SDG 6 (Clean Water and Sanitation), SDG 7 (Affordable and Clean Energy), SDG 9 (Industry, Innovation and Infrastructure), and SDG 11 (Sustainable Cities and Communities). For each goal, the section describes the relevance of the problem, the contribution of the proposed system, and the empirical evidence that will be produced during the project.

The contribution of the project to each goal is stated at its appropriate level. SDG 6, SDG 9, and SDG 11 are primary goals, since the prototype directly treats wastewater, provides infrastructure, and reduces runoff. SDG 1 and SDG 7 are supporting goals, to which the project contributes through its cost, resilience, and passive energy choices.

4.2 Summary Matrix of Sustainable Development Goal Contributions

SDG	Contribution	What AquaPave Does	Evidence to be Produced
SDG 1: No Poverty	Supporting	Provides a low-cost unit built from local materials that lowers a small garage's demand for tap water and its exposure to blocked drains and flooding.	Material cost, water saved per day, and a simple payback estimate (Objectives 4 and 5).
SDG 6: Clean Water and Sanitation	Primary	Treats wash water on-site through a permeable pad, filter layers, and oil separation, and stores it for non-potable reuse.	Influent and effluent comparison, removal efficiency, comparison with reuse targets, and water saved (Objectives 1, 3, and 4).
SDG 7: Affordable and Clean Energy	Supporting	Uses gravity-driven treatment with no pump and a low-power sensor node that can run on a small solar supply.	Number of pumps (none in the prototype) and measured power of the sensor node (Objectives 2 and 5).
SDG 9: Industry, Innovation and Infrastructure	Primary	Combines pavement, treatment, and reuse in one locally buildable unit designed across seven engineering disciplines.	Leak-tested prototype, materials list, construction steps, clogging and cleaning results (Objectives 2 and 5).

SDG	Contribution	What AquaPave Does	Evidence to be Produced
SDG 11: Sustainable Cities and Communities	Primary	Allows water to soak in, reducing runoff and the load on drains, and keeps oily sludge out of the drainage network.	Runoff coefficient compared with a dense control slab and drain-time record (Objective 3).

4.3 Detailed Compliance Assessment by SDG

SDG 1: No Poverty (Supporting)

- **Relevance:** Water scarcity and urban flooding place a heavier burden on small businesses and low-income households, which have the least capacity to absorb additional costs and disruption. Small garages and car-wash operators are among the water users affected.
- **Contribution of AquaPave:** The system is intended as a low-cost unit that can be built from locally available materials and maintained without specialized equipment, making it accessible to small enterprises. Reusing treated water reduces dependence on purchased or rationed tap water, and keeping oily sludge out of drains reduces the risk of blockages and flooding that damage property and interrupt livelihoods.
- **Evidence:** The feasibility assessment (Objective 5) will report the material cost of the prototype, any additional cost relative to conventional alternatives, and an estimate of the cost recovered through water savings.

SDG 6: Clean Water and Sanitation (Primary)

- **Relevance:** Wastewater from garages and car-wash areas is commonly discharged into drains with little or no treatment. This pollutes drains and rivers and wastes water in a city whose supply is already limited.
- **Contribution of AquaPave:** The system forms a complete treatment-and-reuse chain: wash water passes through the permeable slab and filter layers, is collected by the underdrain, is separated from free oil, and is stored for non-potable reuse—reducing pollution at the source and lowering fresh water demand.
- **Evidence:** The prototype will be tested with synthetic wash water and compared with a dense control slab, with samples taken at each stage of treatment and each test repeated at least three times against the reuse targets in Section 3.3.1.

SDG 7: Affordable and Clean Energy (Supporting)

- **Relevance:** Treatment systems that depend on electricity for pumping or dosing are costly to operate and vulnerable to power interruptions.
- **Contribution of AquaPave:** Filtration and oil separation are passive and gravity-driven, requiring no electrical energy. The monitoring node uses low-power components capable of running on a small solar panel and battery (assessed by Electrical Engineering). Reusing water also indirectly reduces the

energy needed by the utility to supply municipal water (noted qualitatively). Where a full-scale unit supplying a wash hose requires a small pump, this will be stated.

- **Evidence:** The report will record the number of pumps used in the prototype (zero) and the measured power consumption of the sensor node.

SDG 9: Industry, Innovation and Infrastructure (Primary)

- **Relevance:** There is currently no low-cost, locally buildable system that combines a permeable wash pad, oil separation, filtration, and reuse storage with performance monitoring (Section 1.2.1).
- **Contribution of AquaPave:** Proposes one integrated pavement-based unit in place of separate pads, traps, and tanks, co-designed by seven engineering disciplines using locally available materials and keeping permeable pavers as a backup.
- **Evidence:** Objective 2 requires a prototype that passes a leak test and directs water through the slab; Objective 5 requires a written assessment of cost, materials, construction steps, clogging behavior, and cleaning.

SDG 11: Sustainable Cities and Communities (Primary)

- **Relevance:** The growth of paved surfaces in Addis Ababa increases surface runoff, and garage waste discharged into drains reduces capacity and contributes to local flooding (Section 3.1).
- **Contribution of AquaPave:** The permeable wash pad allows water to soak into the sub-base instead of running off a sealed surface, while the oil separation chamber and removable filter media keep oil and sludge out of the drainage network.
- **Evidence:** The prototype and dense control slab will receive identical applied flow to compare runoff coefficients, and the clogging test will record drain-time changes and post-cleaning recovery (Section 1.4.7).

4.4 Traceability Matrix: Objectives, SDG Targets, Evidence & Timeline

Every specific objective contributes to at least one of the selected SDGs. The table below links each objective to the SDGs it serves, the evidence to be produced, and the date by which that evidence is available (indicators marked 'proposed' are additions to Section 1.3.2 to be confirmed with the advisor after initial trials):

Objective	SDGs Served	Evidence (Success Indicator)	Available By
1. Characterize the wastewater	SDG 6	Synthetic wash water close to typical published values.	Progress report, 4 Dec 2026
2. Design and build the prototype	SDG 7, 9, 11	Leak-tested prototype with flow through the slab; no pump used in the prototype (proposed).	Prototype fabrication evaluation, late Dec 2026
3. Evaluate treatment efficiency	SDG 6, 11	At least 80% turbidity reduction (proposed), no visible oil film, and much less runoff than the control slab.	Prototype demonstration, 30 Dec 2026
4. Assess reuse potential	SDG 1, 6	Statement of suitable reuses against published limits, water saved per garage per day, and estimated value of water saved (proposed).	Final report, 1 Jan 2027
5. Assess feasibility	SDG 1, 7, 9, 11	Cost, materials, construction steps, clogging and cleaning recovery; payback estimate (proposed); measured power of the sensor node (proposed).	Final report, 1 Jan 2027

4.5 Scope Boundaries, Limitations, and Section 4 References

- **Bench-Scale Prototype Scope:** AquaPave is a single bench-scale prototype. It demonstrates how the design can contribute to the SDGs; it does not by itself change city-wide water, flood, or energy conditions.
- **Indicative Reuse Benchmarking:** The numerical targets are goals to be tested. The comparison with reuse limits is indicative only and is not a certified standard, and the treated water is intended for non-potable reuse only.
- **Tiered Contribution Attribution:** The contributions to SDG 6, SDG 9, and SDG 11 are primary, while SDG 1 and SDG 7 are supporting.

1.5 Section 1 References

The following peer-reviewed articles, theses, engineering standards, and regulatory documents underpin the problem formulation, hydraulic sizing, and experimental methodology presented in Section 1:

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